NGUYEN GIA BACH

Da Nang, Vietnam | gbachnguyen@gmail.com | +84 795 551 811 | Portfolio | GitHub | LinkedIn

EDUCATION

Le Quy Don High School for the Gifted **Da Nang, Vietnam**

Expected Graduation: May 2027

GPA: 9.5/10.0 (Grade 11) | **SAT:** 1410 | **IELTS Academic:** 7.5

Selected Honor: Second Prize in Physics, Da Nang City Excellent Student Competition, 2024

SELECTED ENGINEERING PROJECTS

eSight | *Independent Engineering Project* **Ongoing**

Project Lead

- Developing a smart cane that combines ultrasonic sensing and computer vision to help people with visual impairments detect nearby obstacles.
- Integrated Arduino control logic with graduated buzzer feedback, translating distance and risk into slow, rapid, or continuous alerts.
- Led iterative prototyping, testing, documentation, and presentation; earned Third Prize in the engineering category of the school Science and Engineering Competition.

JWST Space Explorer | *NASA Space Apps Challenge 2025* **October 2025**

Team Lead and Developer

- Led scope and delivery during a 48-hour hackathon and developed a Python/FastAPI workflow to query public JWST observations through NASA MAST.
- Processed FITS imagery with Astroquery, Astropy, and pyvips into Deep Zoom tiles for interactive inspection with React and OpenSeadragon.
- Integrated a Gemini assistant for guided analysis; advanced from local judging to the Global Judging stage as a 2025 Global Nominee.

AstroVerse | *Astronomy Learning Platform* **2026**

Product Developer

- Built a React, TypeScript, and Vite astronomy environment with an interactive sky map, source-attributed space data, and an exoplanet habitability lab.
- Connected observation, physical modelling, spaced-repetition review, and a private science journal in one learning workflow.

ENGINEERING EXPERIENCE

D&D Investment & Technology Development JSC **Da Nang, Vietnam**

Engineering Intern | *ENSURE Da Nang* February - August 2026

- Observed and assisted with documentation of selected equipment-installation activities for ENSURE Da Nang under technical supervision.
- Compiled, entered, and cross-checked deployment information for installation-monitoring records.
- Participated in project discussions and learned how monitoring equipment supports data collection, transmission, storage, and operational monitoring.

LEADERSHIP AND HONORS

RMIT Tech Camp | *Team Lead, Team LQD Da Nang* **2026**

- Led the team to Second Place in the Grand Finals.

FAST Maze Runner | *Robotics Team Lead, The Physicians* **April 2025**

- Led the team to Third Prize in the Robotics Maze Solver Challenge.

Nguyet Du Charity Campaign | *Financial Manager* **February 2026**

- Managed fund allocation for VND 4.2 million and 42 care packages for revolutionary veterans in Da Nang.